An Analysis of Government of Canada Contract Spending (2017–2026)

1. Introduction

1.1. Executive Summary

Government procurement is often viewed through the lens of initial acquisitions the publicized purchasing of fleets, infrastructure, and standardized goods. However, a deep dive into the federal contract registry reveals a different reality. Modern Government of Canada (GoC) procurement has transformed into an **intangible economy**, where the long-term cost of operating, supporting, and extending assets far outweighs the initial cost of buying them.

Between 2017 and 2025, federal procurement experienced a radical structural pivot. While new project starts dominated the landscape prior to 2022, the system has since been consumed by an "Amendment Surge." By 2025, contract amendments officially outnumbered new contract awards. This shift reveals a massive, hidden financial footprint: capital is increasingly concentrated into existing, complex multi-year projects driven heavily by the National Defence and IT service sectors.

1.2. Objective and Scope

The objective of this report is to move beyond superficial reporting metrics and calculate the *true* year-over-year financial footprint of federal contracts. To achieve this, we engineered an advanced analytical pipeline to process a bulk export of federal contract records

In this analysis we seek to answer the following questions:

- **Where is the money flowing?** Breaking down spending by department, threshold, and commodity type to identify true cost drivers.
- **How is the money being awarded?** Evaluating the shift toward Open Bidding (OB) for high-value projects versus the reliance on Traditional Competitive (TC) methods for high-volume operational needs.
- **What is the lifecycle of a modern federal contract?** Tracing how initial awards balloon over time through our custom "Linear Spreading" and amendment tracking algorithms.

1.3. Challenges faced in dataset

Analyzing open government financial data presents significant systemic challenges. The raw data is highly transactional, noisy vendor names, logical chronological errors, and overlapping duplicate entries. Without resolving these issues, standard analyses vastly overcount project volumes and misrepresent the timeline of government spending.

To build a reliable foundation for our insights, we implement a rigorous data engineering methodology utilizing Natural Language Processing (NLP) for vendor entity resolution and multi-layered deduplication logic to transform over a million noisy administrative records into a clean, multi-year spending timeline.

2. Data Engineering & Methodology

Analyzing millions of transactional government records requires a robust data engineering pipeline to transition from raw operational logs to a clean, analytical financial model. The following multi-stage methodology was applied to ensure data integrity and structural accuracy.

2.1. Data Ingestion & Dimensionality Reduction

The foundational dataset is a bulk export of federal contract records consisting of over one million individual entries and 43 distinct variables. An Excel-based ingestion format was purposefully selected over CSV to preserve the stability of extended metadata fields (such as text comments), which are frequently malformed by encoding issues in standard open-data formats.

Upon ingestion, the dataset presented a significant memory burden. To optimize processing speed for subsequent Natural Language Processing (NLP) and iterative spreading algorithms, a strict dimensionality reduction was executed. So, we dropped 24 non-essential columns, resulting in a 53% reduction in memory footprint. Eliminated columns included:

- A**dministrative Metadata:** Fields detailing the mechanical nature of the award (e.g., agreement_type_code, standing_offer_number), which do not impact aggregate financial values.

- **Bilingual Redundancy:** French descriptive fields (description_fr, comments_fr) were purged. Because the subsequent text normalization engine is optimized for English, retaining bilingual equivalents would double memory usage without providing unique analytical variance.

- **Low-Density Indicators:** Socioeconomic tags with less than 1% non-null completion rates were removed to prevent extreme model sparsity

To ensure grouping integrity later in the pipeline, remaining text fields comments_en , owner_org_title were forced to uppercase and stripped of whitespace to prevent string mismatches resulting from basic casting typos.

2.2. Chronological Normalization

Government financial data frequently contains logical timeline errors. A specialized chronological cleaning matrix was developed to resolve "Impossible Dates":

- **Anticipatory Awards:** Contracts logged with a start_year preceding the legal contract_date were capped, forcing the start year to match the legal signing year.
- **The "Indefinite" Problem:** Thousands of entries possessed a start_year but lacked a defined delivery_date (end year). Rather than dropping these records which would under-report total spending by millions—the model assumes a default one-year duration by filling null end dates with the start year.
- **Future-Date Capping:** Any end_year extending beyond 2029 (including obvious data entry errors like "9999" or "3103") was truncated back to the start_year, treating it as a single-year operational expense rather than spreading it across centuries

2.3. Vendor Name Normalization

A primary challenge in procurement analytics is corporate fragmentation; identical entities appear under varied names due to inconsistent regional tagging, bilingual entries, and historical mergers.

To achieve an accurate view of vendor consolidation, a two-layered Entity Resolution engine was deployed:

- **Deterministic Master Mapping:** The pipeline initially checks raw vendor names against a curated vendor_comparison_master.csv file. This high-speed, static mapping provides absolute "ground truth" for the largest federal vendors, preventing the conceptual "drifting" of identities that unsupervised algorithms occasionally cause.

- **NLP Fuzzy Matching:** For the "long tail" of unmapped vendors, the pipeline utilizes the scikit-learn library. A regex filter first strips generic industry noise words (e.g., "TECHNOLOGIES", "CONSULTING") to isolate the core corporate identifier. The names are then processed through a TfidfVectorizer utilizing character-level trigrams. By breaking words into 3-character chunks, the algorithm successfully matches variations despite typos or truncated legal suffixes.

- **Optimized Blocking:** To compute similarities efficiently, vendors are only compared if they share the first 5 characters of their normalized name. Matches are merged based on a strict 0.78 similarity threshold, balancing variation capture against false-positive collisions

2.4 The Deduplication & Amendment Matrix

Because the open dataset is an unlinked transactional ledger, an original $50,000 contract and its subsequent $10,000 amendment appear as two completely distinct rows. Furthermore, redundant data publication creates pure duplicate rows.

To distinguish between *new spending* (amendments) and *redundant data* (duplicates), a strict methodology was established:

- **Strict Subset Deduplication:** Rows are flagged as duplicates only if they share a perfect match across six exact variables: Procurement ID, Department, Master Name, Contract Date, Contract Value, and Original Value. This rigorous criterion retains actual financial changes while purging administrative redundancies.

- **Composite Key Grouping:** To link a base contract to its future amendments, a composite primary key (contract_id) was engineered by grouping rows matching the same procurement_id, start_year, and owner_org.

- **Amendment Extraction:** Within these groups, rows where amendment_value is non-null and non-zero receive an is_amendment = 1 flag. Crucially, the pipeline uses regular expressions to extract the precise amend_year from the fiscal reporting period, allowing the model to know exactly *when* new capital was injected into the project lifecycle.

2.5 Normalization by Year (The Linear Spread)

To visualize true annual budget commitments, the total value of each contract must be distributed across its active timeline.

$$Yearly\ Rate\ = \backslash frac\{Total\backslash\ Contract\backslash\ Value - Cumulative\backslash\ Amount\backslash\ Already\backslash\ Spent\}\{Years\backslash\ Remaining\backslash\ in\backslash\ the\backslash\ Contract\backslash\ Timeline\}$$

The model relies on a **Linear Spending** assumption, distributing funds evenly across the contract's life. However, instead of a simple division, the algorithm actively integrates amendments by recalculating the rate at every financial segment.

To understand how the algorithm handles overlapping information from amendments, consider a single contract group with four distinct entries published at different times:

- Entry #1 (Initial): Published: 2005. Timeline: 2005–2009. Value: $10M.

- Entry #2 (Amendment): Published: 2007. Timeline: 2005–2011. Value: $10M.
- Entry #3 (Amendment): Published: 2010. Timeline: 2005–2011. Value: $15M.
- Entry #4 (Amendment): Published: 2011. Timeline: 2005–2015. Value: $20M.

The algorithm determines the Effective Entry for each year by selecting the most recent data available at that specific point in time.

2.6 Limitations

Despite aggressive programmatic cleaning, certain systemic limitations remain inherent to the dataset:

- **Structural Blurring:** Complex corporate entities, particularly Joint Ventures and multi-sector parent companies, are merged under single identities, making granular per-industry spending calculations challenging.
- **Threshold Collisions:** Because federal departments frequently award contracts at precise sole-source caps (e.g., exactly $25,000), legitimate multi-award days for the same vendor can occasionally trigger false positives in the strict deduplication engine.

3. Analysis (2017–2026)

By applying the linear spreading algorithm and amendment grouping to the cleaned dataset, a distinct narrative emerges: the Government of Canada is transitioning out of an era of new acquisitions and into an era of heavy, compounding maintenance.

3.1 Total government-wide contract and amendment entries

Between 2017 and 2025, the overall volume of federal procurement remained remarkably resilient, anchored by a highly stable influx of new project starts. Rather than a system experiencing radical structural flips, the data reveals a steady, consistent operation.

Initial contracts consistently formed the absolute majority of procurement activity every single year. The government consistently processed roughly 47,000 to 54,000 new awards annually, proving the baseline acquisition engine kept running smoothly even through global disruptions.

The most striking revelation in the volume data is the trajectory of contract modifications. Contrary to expectations of increasing administrative bloat, the *volume* of contract amendments trended downward over the latter half of the decade. After hitting a peak of roughly 16,000 amendments in 2019, amendment volume steadily decreased, dropping to 8,000 in 2024 and approximately 5,000 by 2025. (Note*: since this data was taken in March 2026 and it gets updated daily; 2025 stands as the final reliable volume benchmark)*

This drop in amendment volume creates a fascinating divergence when paired with the financial data. As we will see in the following section, total government spending exploded to record highs in 2024. If the absolute number of amendments is dropping by half, but total spending is skyrocketing, it suggests a profound shift in project scale. The government is executing fewer extensions overall, but the ones they do execute are massive, high-value capital injections into select "mega-projects." The administrative workload in sheer volume has decreased, but the financial gravity of those individual actions has never been higher.

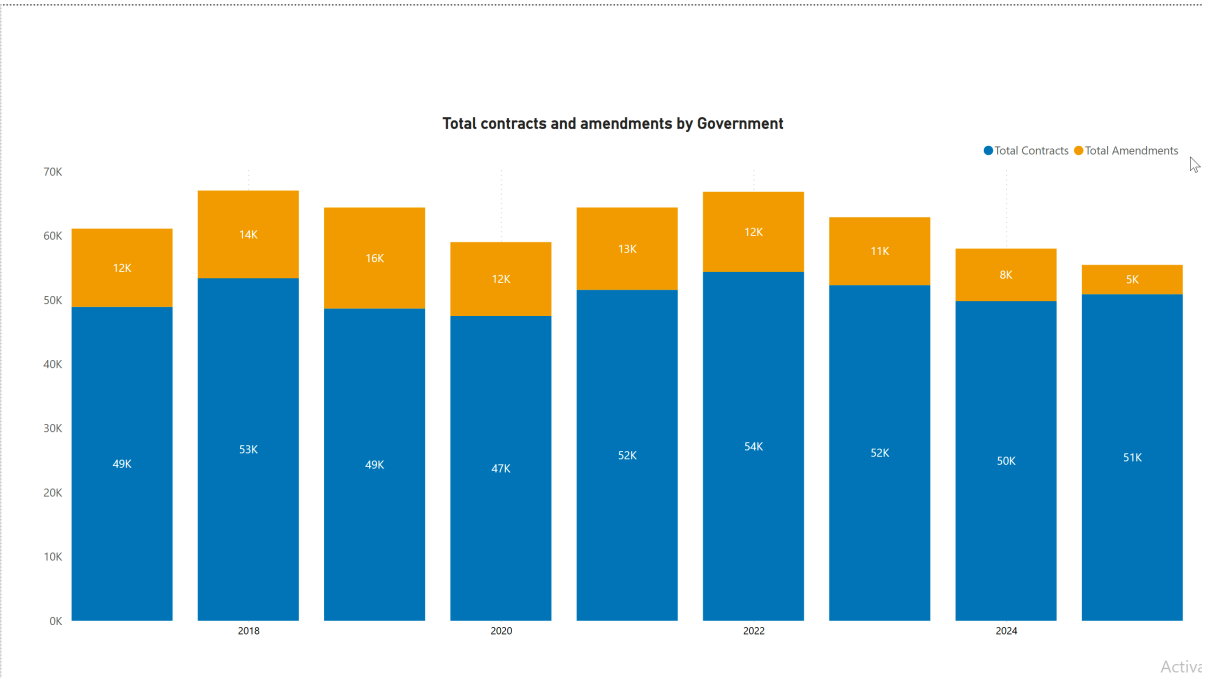

Figure 1: Total Contracts and Amendments by Government (Data Link)

Same data reported in Fiscal Format

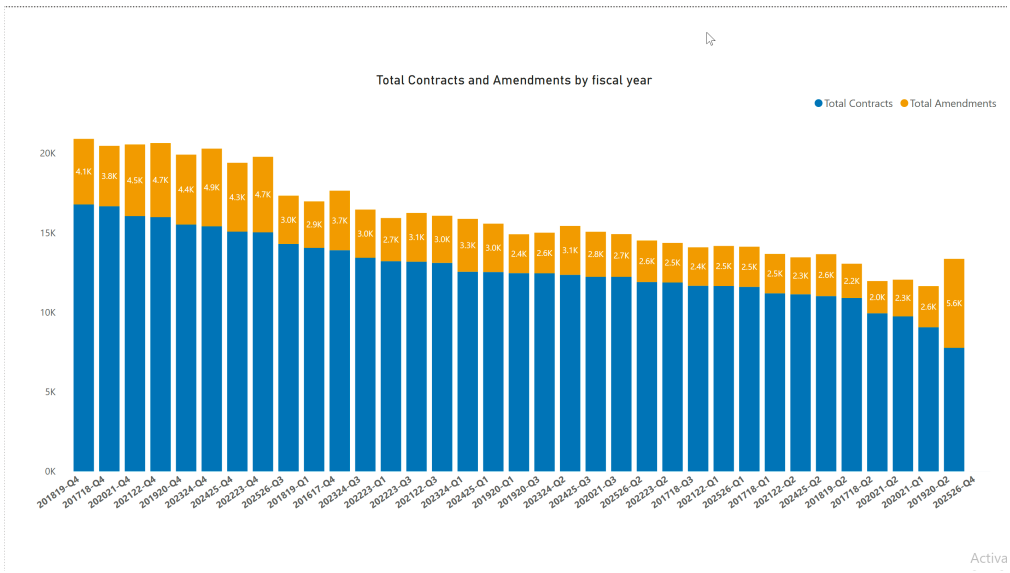

Figure 2: Total contracts and amendments by fiscal quarter (Data Link)

3.2 The 2024 Spending Peak

The shift toward amendments is not merely a matter of extended timelines; it is driven by massive infusions of new capital. The financial model exposes a surge, with total active spending escalating from a $4.8 billion baseline in 2017 to an explosive peak of $37.85 billion in 2024.

Because our linear spreading logic isolates spending strictly for contracts originating in 2017 or later, this chart illustrates a "snowball effect.". The 2024 peak represents the cumulative financial weight of eight years of overlapping, multi-year projects reaching maturity simultaneously. Even factoring in the 2026 reporting lag, the government has already committed $10.3 billion for 2026 based entirely on the future dated liabilities of existing contracts,

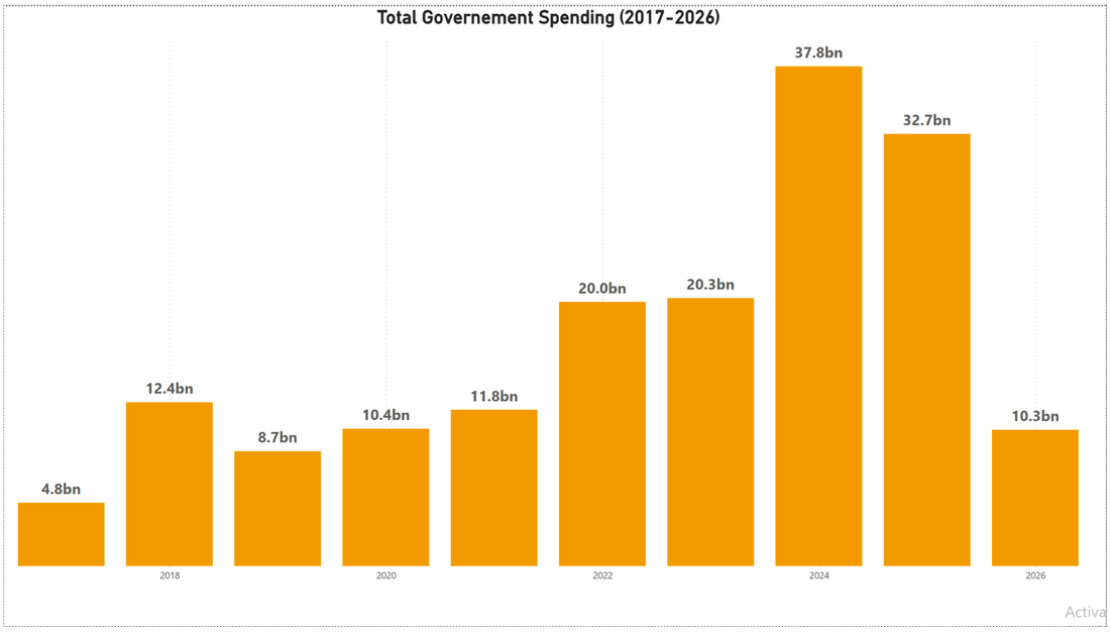

Figure 3: Government Spending (Data Link)

3.3 Departmental Concentration: The DND Surge

This spending surge is not distributed evenly across the federal government; it is overwhelmingly driven by National Defence (DND). Over the ten-year scope, DND's total estimated commitment reached a staggering $100.8 billion nearly five times greater than the second-highest spender, Fisheries and Oceans Canada ($20.7 billion).

During the 2024 peak, DND spending hit $30.6 billion, accounting for nearly 80% of all government-wide contract spending for that year.

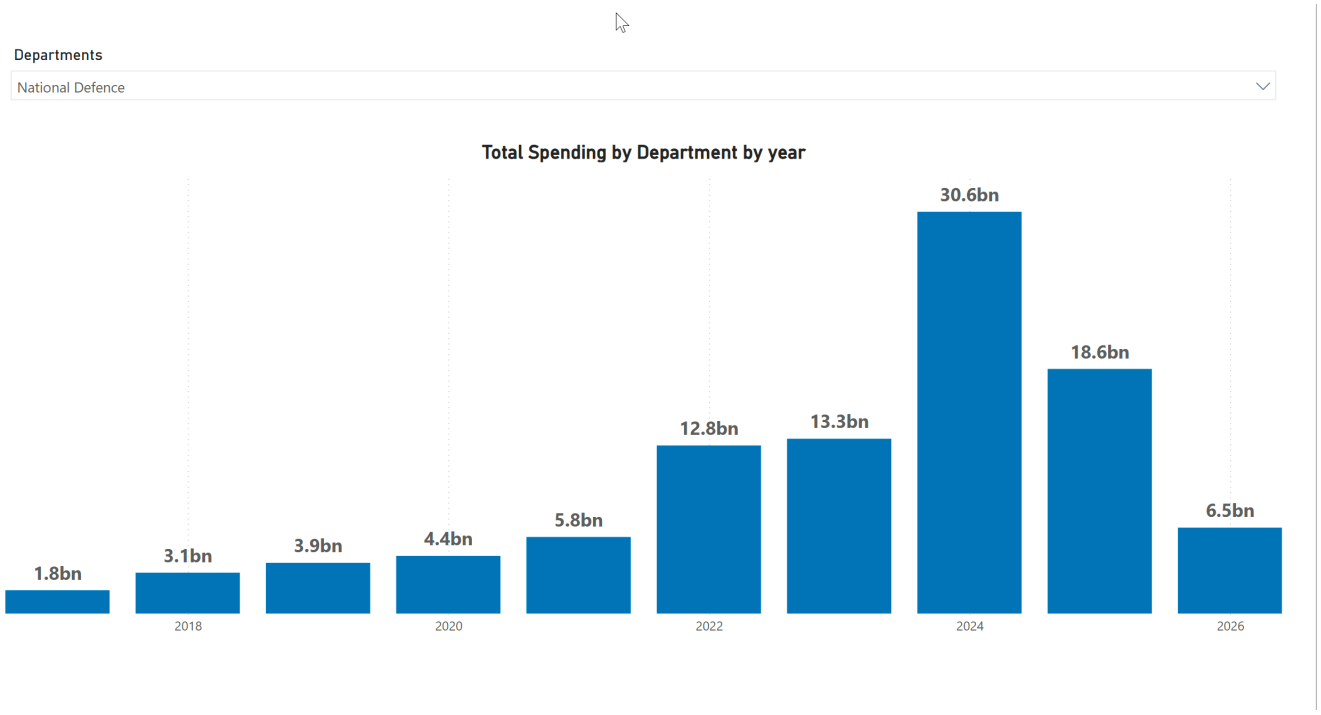

Figure 4: Department wise spending (National Defense Department) [Data Link]

High-Volume vs. High-Value Portfolios: An analysis of the 457,131 initial contracts awarded between 2017 and 2025 reveals a near-perfect operational split across the government: 51.6% were purchases below the $25,000 threshold, while 48.4% were strategic investments exceeding it.

- **The Heavyweights:** DND drives the highest absolute volume (140,215 initial actions), with over half (74,319) falling under $25k, indicating a massive logistical footprint requiring constant low-level operational maintenance.

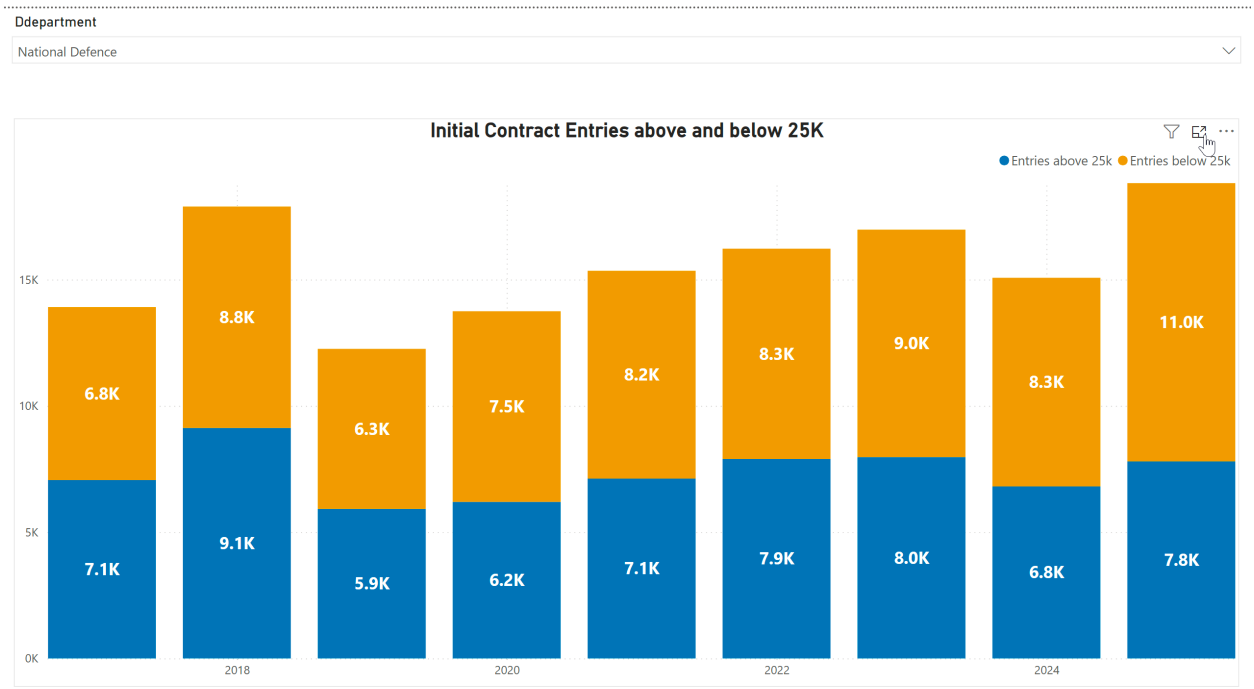

Figure 4: DND department entries

- **The Agile Departments:** Conversely, departments like Global Affairs Canada skew heavily toward high-frequency, low-value actions (17,890 small contracts vs. 13,398 large), suggesting a reliance on quick-turnaround, localized service

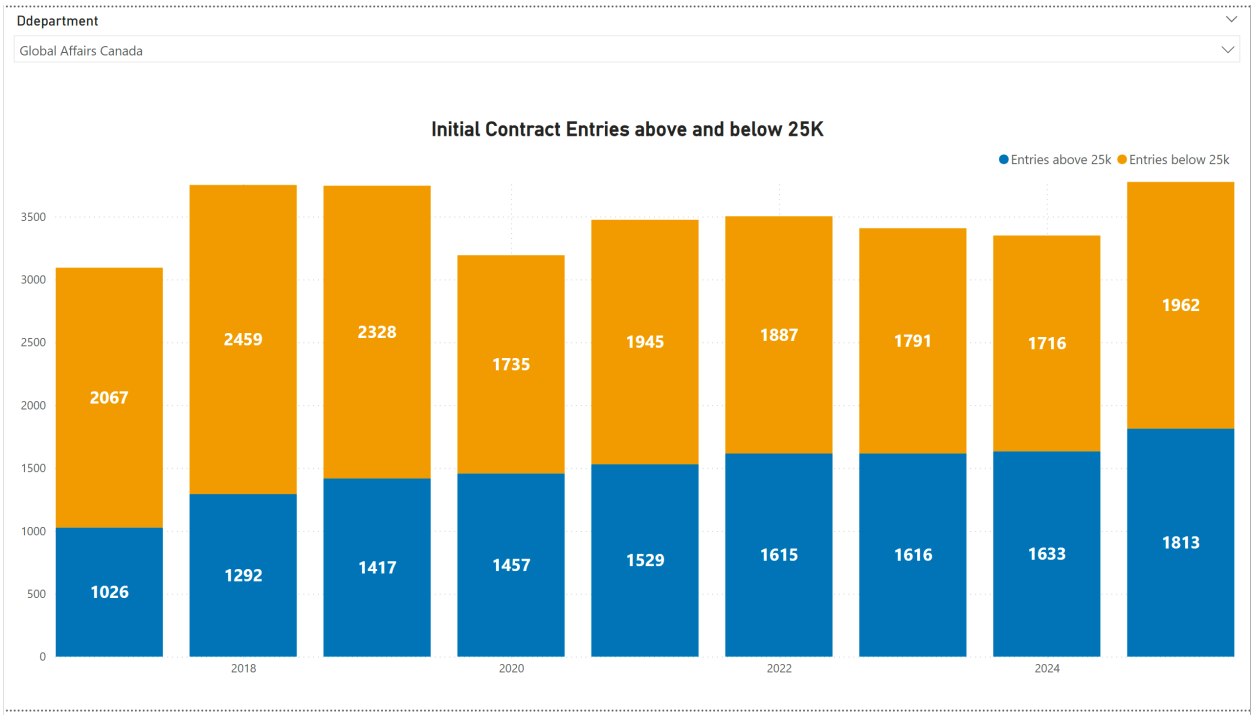

Figure 5: Global Affairs Canada contract entries (Data Link)

3.4 The Maintenance Gap

Breaking down spending by commodity type provides the clearest explanation for the amendment surge: modern federal procurement is fundamentally an intangible, service-based economy.

Across all departments, spending on Services ($107.5B) more than doubled the spending on physical Goods ($44.1B). During the 2024 financial peak, a remarkable 82% ($31.2 billion) of the capital was allocated exclusively to Services.

The Defence Maintenance Gap > While DND is highly visible as a buyer of physical equipment ($31.9 billion on Goods), it spent twice as much on Services ($63.6 billion) over the same period. The long-term cost of operating, supporting, and upgrading a physical asset (IT consulting, specialized maintenance, software) far outweighs the initial cost of acquiring it. Services are inherently more prone to "scope creep" than physical goods, perfectly explaining the systemic shift toward amendment-heavy lifecycles.

Notably, Fisheries and Oceans Canada breaks this trend. Driven by maritime infrastructure and shipyard mandates, Construction ($9.2 billion) is their leading commodity category, surpassing both Services and Goods

Department
All

Year
2024

commodity_type	Total_contracts	% of Yearly Contracts	Total_spending	% of Yearly Spending
S	45K	56.29%	31,185.1M	82.39%
G	30K	37.51%	5,495.9M	14.52%
C	5K	6.20%	1,168.6M	3.09%
Total	**79K**	**100.00%**	**37,849.6M**	**100.00%**

Figure 6: Commodity type breakdown by departments and year (Data Link)

3.5 Solicitation Mechanics: Open Bidding Takes the Lead

The methodology by which funds are awarded has also evolved to match the complexity of these mega-projects. Historically, Traditional Competitive (TC) methods have been the government's workhorse, processing the highest volume (324,552 total contracts) to manage mid-sized operational needs.

Departments: All

Year: All

solicitation_procedure	Sum of total_contracts	% of Yearly Contracts_soliciation	Sum of total_spending	% of Yearly Spending_solicitation
TC	325K	46.70%	63,827.3M	37.69%
OB	120K	17.29%	60,718.4M	35.85%
TN	218K	31.35%	29,561.7M	17.46%
AC	8K	1.09%	8,584.5M	5.07%
ST	24K	3.46%	6,548.7M	3.87%
NULL	1K	0.11%	111.9M	0.07%
Total	**695K**	**100.00%**	**169,352.6M**	**100.00%**

Figure 7: Contracts Awarded by solicitation procedure

However, during the 2024 spending peak, Open Bidding (OB) became the dominant financial mechanism. In 2024 alone, $21.6 billion (57% of the total peak spend) was awarded through Open Bidding, indicating that the highest-value, highest-complexity projects are being pushed to open market competitions.

Conversely, highly specialized non-competitive methods are utilized for extreme-value sole-source innovations. Advanced Contract Award Notices (AC), for instance, handled the lowest volume (7,547 contracts) but commanded a disproportionately massive financial footprint of $8.5 billion

Departments: All

Year: 2024

solicitation_procedure	Sum of total_contracts	% of Yearly Contracts_soliciation	Sum of total_spending	% of Yearly Spending_solicitation
OB	12K	15.07%	21,646.5M	57.19%
TN	27K	33.53%	6,079.3M	16.06%
TC	35K	44.12%	5,645.4M	14.92%
AC	1K	1.11%	3,203.6M	8.46%
ST	5K	6.17%	1,274.7M	3.37%
NULL	0K	0.00%	0.1M	0.00%
Total	**79K**	**100.00%**	**37,849.6M**	**100.00%**

Figure 8: Open Bidding procedure becomes dominant in 2024. (Data Link)

4. Conclusion

This analysis of the Government of Canada's contracts from 2017 to 2026 reveals a procurement system undergoing a profound financial evolution. While the sheer volume of new administrative actions remains incredibly stable year-over-year, the financial gravity of those actions has fundamentally changed.

The analysis definitively highlights a "Mega-Amendment Shift." The absolute volume of contract amendments has trended downward since 2019, yet total estimated contract spending exploded to an unprecedented peak of $37.8 billion in 2024. This divergence proves that capital is increasingly being concentrated into a smaller number of highly complex, incredibly expensive, multi-year projects.

Furthermore, this financial snowball is not evenly distributed. It is distinctly driven by an "Intangible Economy" where Services account for $107.5 billion of the total spend, dwarfing the acquisition of physical Goods. Anchored by the Department of National Defence which committed a staggering $100.8 billion over the analyzed period the government is clearly facing a "Maintenance Gap." The long-term, service-based costs of operating, supporting, and extending federal assets now represent the dominant force in public procurement.

5. Strategic Recommendations

Based on the insights derived from this dataset, the following actionable recommendations are proposed for procurement oversight and strategic planning:

- Procurement oversight currently places heavy emphasis on the initial award value and the volume of Traditional Competitive contracts. Given that the highest financial risks and capital concentrations now occur during the amendment phase of existing contracts, oversight bodies must shift to tracking "Total Value Over Time."
- Departments with heavy capital acquisitions (such as DND and Fisheries and Oceans) must adapt their fiscal planning to reflect the reality of the intangible economy. Strategic budgets should heavily front-load anticipated service, IT, and operational maintenance costs, rather than treating them as subsequent ad-hoc amendments.
- The current open-data standard publishes transactional snapshots rather than relational databases. To improve public transparency and analytical accuracy, the Government of Canada should natively link amendments to their base contracts using a persistent, unique primary key across all fiscal years, eliminating the need for external analysts to reverse engineer composite keys.

6. Future Work & Next Steps

This analysis establishes a clean, normalized baseline for federal contract data. To further expand on these findings, future work should focus on:

- **Advanced Vendor Entity Resolution:** The current NLP pipeline successfully normalized the most frequent vendor aliases to resolve massive corporate fragmentation. However, given time constraints, the "long tail" of smaller vendors requires more exhaustive cleaning. Future deep-dive analyses at the vendor level must systematically untangle complex corporate hierarchies, joint ventures, and historical mergers to provide an absolute measure of vendor concentration and monopoly power.
- **Automated Vendor Industry Determination:** At present, conducting an industry-by-industry breakdown of federal contracts requires manually grouping vendors. Future iterations should automatically enrich the dataset by linking normalized vendor names to external data sources (e.g., corporate registries, Wikipedia APIs) to identify their primary industry. This would enable government-wide aggregate statistics by market sector, rather than simply highlighting the largest vendors in a manually curated field.
- **Contract-by-Contract Machine Learning Classification:** Aggregate statistics by industry are currently much easier to complete at the company level. However, many of the largest vendors contracting with the Canadian government (e.g., telecommunications giants) provide both physical hardware and IT consulting services. Future work should utilize the existing economic_object_code and description_en fields to train a machine learning classifier, allowing us to consistently determine the specific industry or nature of work on a granular, *per-contract* basis.
- **Granular Date Parsing and Fiscal Year Alignment:** Given the chronological formatting issues and data quality limitations in the raw open data, the current Linear Spreading algorithm calculates the effective timeline to the nearest calendar year. By implementing more advanced date parsing algorithms and developing programmatic assumptions for missing month and day values effective spending could be calculated using exact daily burn rates. This upgrade would allow all aggregate statistics to map perfectly to the Government of Canada's official Fiscal Year (April 1 to March 31), allowing for a 1:1 comparison with official Public Accounts data.